THE PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
NAMANGA SECONDARY SCHOOL
FORM FOUR PRE-MOCK EXAMINATION
033 BIOLOGY

Time: 3 Hours **May 2026**

Instructions

1. This paper consists of sections **A**, **B** and **C** with a total of eleven questions.
2. Answer all questions in sections **A**, **B**, and two questions from section **C**.
3. All writing should be in blue ink, except for diagrams, which must be drawn in pencil.
4. Cellular phones and any unauthorized materials are not allowed in the examination room.
5. Write your Examination Number on every page of your answer sheets.

SECTION A (16 MARKS)

Answer all questions in this section

1. Choose the most correct answer among the given alternatives and write down the answer in the space given.

   (i) Which is a characteristic of all living organisms?
   - A. Photosynthesis
   - B. Locomotion,
   - C. Irritability,
   - D. Phagocytosis.
   - E. Flying

   (ii) A group of organisms that can interbreed and produce fertile offspring is called:
   - A. Kingdom
   - B. Species
   - C. Phylum
   - D. Family.
   - E. Genus

   (iii) Food moves down the oesophagus by a wave-like motion called….
   - A. Asphyxiation
   - B. Convolutions
   - C. Peristalsis
   - D. Suffocation
   - E. Coagulation

   (iv) Which disease can be controlled or prevented by boiling drinking water, washing hands before eating and eating warm foods?
   - A. Cancer
   - B. Tuberculosis
   - C. Rickets
   - D. Bilharzia
   - E. Cholera

(v) At puberty, Neema breasts enlarged and after getting married and getting a baby, her breasts began releasing milk for her baby. Which chemicals are responsible for that?

A. Prolactin
B. Oxytocin
C. Testosterone
D. Progesterone
E. Thyroxine

(vi) Genetic disorder caused by failure of X chromosome to separate during the process of egg formation, resulting into an individual with two X chromosome and one Y chromosome.

A. Turner's syndrome
B. Polyploidy
C. Sickle cell anaemia
D. Down's syndrome
E. Klinefeleter syndrome

(vii) The appearance of new characters that has never been before in a population is due to:

A. Mutation
B. Use and disuse
C. Theory of special creation
D. Theory of spontaneous generation
E. Lamarck's theory

(viii) The enzyme that digests proteins in the stomach is:

A. Pepsin
B. Amylase
C. Lipase,
D. Trypsin.
E. Sucrase.

(ix) A person with blood group O is a universal donor because their blood:

A. Lacks antibodies a and b.
B. Has both antigens A and B.
C. Lacks antigens A and B.
D. Has antibodies a and b.
E. Has antigen A only.

(x) Which part of the brain maintains body posture and balance?

A. Medulla oblongata
B. Cerebrum
C. Cerebellum
D. Hypothalamus.
E. Fore brain

2. Match items in list **A** with the correct phrase in list **B**.

LIST A	LIST B
(i) Their cell walls are made up of chitin. (ii) The gametophyte is the dominant generation. (iii) It has a developed conductive tissue while its gametophyte is free living. (iv) An aquatic, unicellular, eukaryotic, photosynthetic organism which moves by means of flagella. (v) Lives in water but comes to lay soft shelled eggs on land. (vi) Possesses four pair of legs, lacks antennae and has bod divided into two main parts.	A. Frog B. Arachnid C. Turtle D. Euglena E. Paramecium F. Fern plant G. Moss plant H. Hippopotamus I. Fungi J. Crustacean

SECTION B (54 MARKS)

Answer all questions in this section.

3. (a) (i) Define 'Biology Laboratory'.
   (ii) Differentiate specimen from model as used in Biology
   (b) Among the laboratory rules posted up on the Seypalm Secondary School notice board by the laboratory entrance are: (i) No eating or drinking. (ii) Wear lab coat and safety goggles. (iii) Label chemical containers. Why is each of three above mentioned rules important?

4. (a) A trophic level is a group of organisms in a food chain. At trophic levels, species are categorized into heterotrophs and two other main groups.
   (i) With an example of an organism in each category, outline the other two main groups species are categorized based on trophic levels.
   (ii) Explain the different functional groups heterotrophs can be further sub-divided into.
   (b) Briefly explain why there are fewer predators than prey.

5. (a) Draw a well-labeled diagram of a human urinary system.
   (b) (i)Explain two functions of the kidney in homeostasis.
   (ii)Give reason(s) why during cold days, individuals produce large and dilute urine.

6. (a) Is it always possible to tell whether a person is a biological child to a father by using blood groups only? Explain.
   (b) Consider a man with blood group A, and his wife of blood group B, having four children with blood groups AB, B, A and O respectively. Using allele I^A as corresponding to blood group's antigen A, allele I^B to blood group's antigen B and allele i to no blood group's antigen:
      (i) Explain how the phenotypes (blood groups) are related to the alleles (genotypes).
      (ii) With the aid of a Punnet square cross, show whether or not the husband could be the biological father to the child with blood group O.
      (iii) Show the expected phenotypic ratio of their offspring.

7. (a) Define 'Transpiration'.
   (b) Explain how temperature, humidity, and wind speed affect transpiration rate.

8. (a) State four functions of the rib cage and three functions of the vertebral column in a mammalian skeleton.
   (b) Differentiate between a tendon and a ligament.

SECTION C (30 MARKS)
Answer two questions from this section.

9. Discuss the human reproductive system, covering: (a) primary male and female sex hormones and their roles, (b) stages of the menstrual cycle, and (c) three importance of family planning in Tanzania.

10. Define variation and briefly explain three causes of genetic variation and conclude by explaining why variation is beneficial to the species but not necessarily to an individual member of the species

11. Discuss waste management challenges in Tanzania urban areas by: explaining four types of waste based on their physical states, discussing six environmental and health problems of poor waste disposal.